Measurement and Causality

Week 2

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Today

1. **Shoe leather.** Freedman on what regression can and cannot do, and John Snow on what it looks like when you do it right.

2. **The toolkit** — at the board. Where bias comes from, and the four ways we try to kill it: randomization, differences, instruments, and honest standard errors.

Last week I made a lot of claims about the past. Today: how would we know?

Shoe leather

Four things people believe about regression

1. Regression usually works, although it is (like anything else) imperfect and may sometimes go wrong.

2. Regression sometimes works in the hands of skillful practitioners, but it isn't suitable for routine use.

3. Regression might work, but it hasn't yet.

4. Regression can't work.

Freedman places himself at **3**, and spends the paper explaining why. Freedman (1991), "Statistical Models and Shoe Leather."

Snow's problem, 1854

- Cholera kills tens of thousands in London. The dominant theory is **miasma** — bad air rising from filth.
- Snow's competing theory: cholera is **waterborne**, transmitted by something in the drinking water that reproduces in the gut.
- No germ theory yet. No microscope evidence. No regression.

Two theories, both consistent with the fact that cholera clusters in poor, filthy districts. **What evidence could separate them?**



Snow's map of the Broad Street outbreak, 1854. Bars are deaths; the labelled circles are public pumps.

The Broad Street pump

- Deaths cluster tightly around **one pump**. Snow persuades the parish to remove its handle.

- But he does not stop there — and this is the part that matters. He chases **the anomalies**:
  - the workhouse in the middle of the outbreak, with its own well, almost untouched;
  - the brewery whose workers drank beer, untouched;
  - a widow in Hampstead, miles away, dead — who had Broad Street water delivered because she liked the taste.

Each anomaly is a place the miasma theory predicts one thing and the water theory predicts another.

Then he finds a natural experiment

Water supplier	Houses	Cholera deaths	Deaths per 10,000
Southwark and Vauxhall	40,046	1,263	**315**
Lambeth	26,107	98	**37**
Rest of London	256,423	1,422	59

- Two companies served **the same streets**, often **neighboring houses**. Households had not chosen their supplier and mostly did not know it.
- In 1852 Lambeth moved its intake **upstream**, above the London sewage discharge. Southwark and Vauxhall did not.

Snow's Table IX, as reproduced in Freedman (1991), Table 1. Freedman (p. 296) dates the Lambeth move to 1849. The new intake opened in 1852.

Why that table is the whole argument

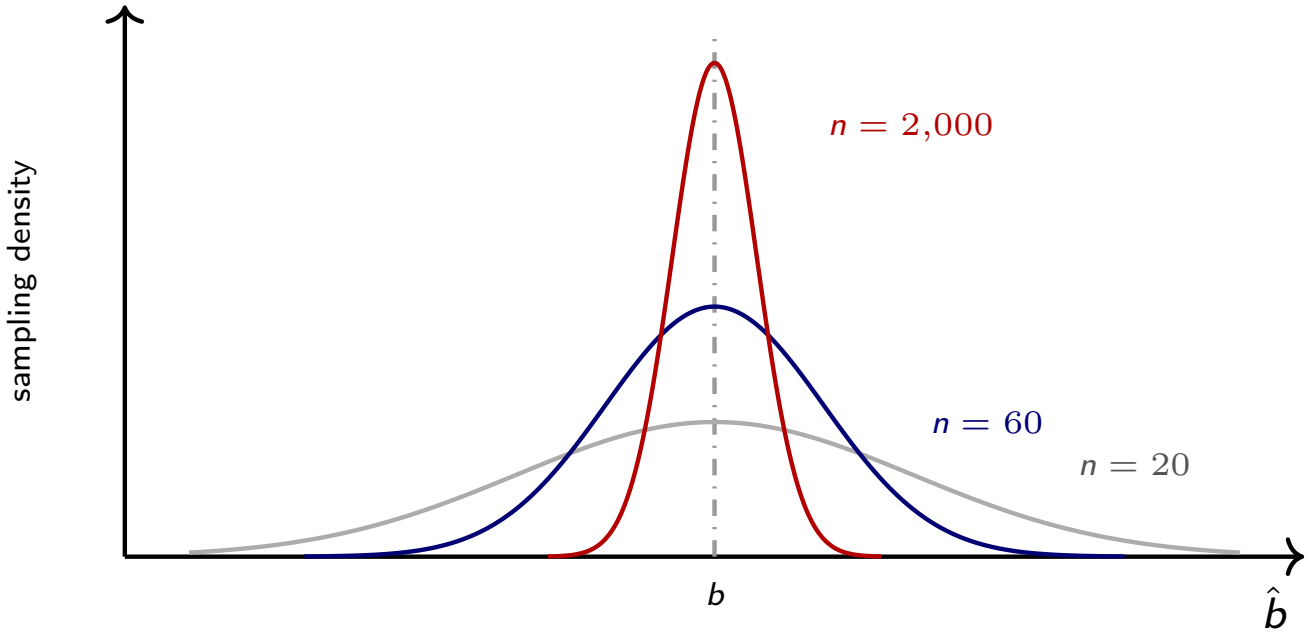
- Assignment to **treatment** (water source) is effectively **as good as random** with respect to everything else — income, occupation, filth, street, air.

- So the eight-fold difference in death rates cannot be explained by the things miasma theory points at. **They are held constant by the design.**

Snow had no germ theory, no statistical model, and no computer. He had a **comparison that was fair by construction** — and he got the mechanism right thirty years before anyone saw the organism.

The toolkit

What "unbiased" and "consistent" actually look like



- **Unbiased**: the distribution is centred on the truth at *any* sample size. All three curves sit over b.
- **Consistent**: it collapses onto the truth as n grows. $\lim_{n\to\infty} \hat{b}(n) = b$.
- **Neither says your particular estimate is close.** That is what the standard error is for.

Before you believe any coefficient

Always look for four things:

1. **Sign.** Is it the direction the theory predicted?
2. **Size.** Is it big enough to matter? Compared with what?
3. **Significance.** Could it plausibly be zero?
4. **Sources of bias.** What would have to be true for this to be causal?

Three of these are arithmetic. The fourth is the job, and it is what I will ask you about in the oral exams.

Next week

- **Week 3 (14 September): Colonialism as a Natural Experiment.**

- Acemoglu, Johnson and Robinson use settler mortality as an instrument — exactly the Z_i from today's board work.
- Dell uses a boundary. Conley and Kelly ask whether any of it survives honest standard errors.

Note the gap: no class on 7 September, Labor Day. A full fortnight between today and Week 3, so start the reading early.